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ORIGINAL ARTICLE

Metabolic acidosis in patients with diabetes 2 undergoing cardiac surgery

The impact of SGLT2 inhibitor use: a retrospective cohort study

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BACKGROUND Sodium-glucose cotransporter 2 inhibitors (SGLT2-i) lower blood sugar and reduce cardiovascular events and kidney failure. However, there have been increasing reports of euglycaemic diabetic ketoacidosis (eDKA) linked to SGLT2-i medicines.

OBJECTIVE Investigating the association between SGLT2-i use and the incidence of metabolic acidosis in patients with type 2 diabetes undergoing cardiac surgery.

DESIGN A retrospective observational cohort study comprising 121 patients, with 38 in the SGLT2-i group and 83 in the control group.

SETTING A 2-year period at Haukeland University Hospital, a tertiary regional hospital in Western Norway.

PATIENTS Patients with type 2 diabetes undergoing cardiac surgery.

INTERVENTIONS Collection of clinical and laboratory data, including acid/base balance parameters, surgery details and SGLT2-i use.

MAIN OUTCOME MEASURES Base excess and anion gap measurements as indicators of ketosis development. A subgroup analysis in patients without renal failure (glomerular filtration rate $> 60\,ml\,min^{-1}\,m^{-2}$).

RESULTS Lower base excess levels and increased anion gaps were observed in the SGLT2-i group compared with controls at various time points postoperatively, with no significant differences in serum lactate levels.

Twelve hours postoperatively, 41% of SGLT2-i patients without renal failure had a base excess -3 $mmol\,l^{-1}$ or less after correction for serum lactate (indicating ketosis) compared with only 8% in the control group ($P < 0.001$). The anion gap was elevated in the SGLT2-i group compared to the control group at 12 h postoperatively ($P = 0.018$).

Multivariable regression analysis identified SGLT2-i use as an independent factor associated with a lower base excess after correction for lactate levels ($P < 0.001$). Cessation of SGLT2-i medication did not correlate with the degree of acidosis.

CONCLUSION While taking SGLT2 inhibitors, diabetic patients undergoing heart surgery are at an increased risk of ketosis and possibly metabolic acidosis. This emphasises the importance of careful observation and effective treatment strategies within this group.

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KEY POINTS

- Sodium-glucose cotransporter 2 inhibitors (SGLT2-i) are associated with a higher risk of metabolic acidosis in patients with type 2 diabetes undergoing cardiac surgery.
- Compared with controls, lower base excess levels and increased anion gaps were observed postoperatively in the SGLT2-i group.
- Only 8% of controls compared with 41% of SGLT2-i patients without renal failure exhibited significant ketosis postoperatively ($P < 0.001$).
- SGLT2-i usage was identified as an independent factor for lower base excess corrected for lactate levels, indicating a higher risk of metabolic acidosis ($P < 0.001$).

Introduction

Sodium-glucose cotransporter 2 inhibitors (SGLT2-i) have gained widespread use due to their blood sugar lowering effect and reduction in the incidence of cardiovascular events and kidney failure.[1–10]

The SGLT2-i drugs empagliflozin and dapagliflozin have also shown beneficial effects on cardiovascular events in patients with heart failure without diabetes, regardless of ejection fraction.[8] In 2021, the European Society of Cardiology guidelines recommended SGLT2-i as treatment for heart failure (Class I, Level A recommendation).[11]

Since their introduction in 2013, there has been a growing number of reports concerning euglycaemic diabetic ketoacidosis (eDKA) associated with SGLT2-i. The FDA warned as early as 2015 that use of SGLT2-i could provoke an 'atypical' DKA, an euglycaemic metabolic acidosis with or without a high anion gap.[12]

eDKA was first described in 1973 by Munro *et al.*[13] in a case series of patients with insulin-dependent diabetes: 37 of 211 patients with diabetic ketoacidosis had a blood glucose level of less than 300 mg dl^{-1} (16.7 mmol l^{-1}).

After a severe case of ketoacidosis following a coronary artery bypass graft, we observed a higher-than-expected incidence of acidosis among patients undergoing heart surgery in our department. This led to further investigation into the issue.[14]

The aim of this investigation was to explore the correlation between the use of SGLT2-i and the occurrence of peri-operative metabolic acidosis, as indicated by a low base excess serving as a proxy for diabetic ketoacidosis, in patients diagnosed with Type 2 diabetes undergoing cardiac surgery.

We hypothesised that the use of SGLT2-i would result in an increased incidence of metabolic acidosis.

Materials and methods

The present study is a retrospective observational cohort study conducted at Haukeland University Hospital, a tertiary regional hospital in Western Norway, serving a population of approximately 1.1 million citizens. The hospital performs around 600 open-heart procedures annually.

Initially, we sought approval from the Regional Committee for Medical and Health Research Ethics (REK) Western Norway, to initiate a study. However, due to a high volume of applications at the REK Western Norway, our request was forwarded to REK South-East Norway. Approval was subsequently granted by REK South-East Norway to access data stored in electronic medical records using specific search terms related to cardiac surgery with extracorporeal circulation and diabetes mellitus Type 2 (DM2). Approval was granted under reference number 219230 on 4 February 2021.

Following this approval, we contacted all relevant patients by mail, providing them with detailed information about the study and informing them of their right to withdraw from participation at any time. Retrospective data from the electronic journal system (DIPS) for the study period from 1 January 2019 to 31 December 2020 were collected.

A total of 122 patients with DM2 who underwent cardiac surgery with extracorporeal circulation during the study period were identified. One patient declined to participate. The final study sample consisted of 121 patients: 38 were on SGLT2-i therapy (SGLT2-i group) and 83 were not taking these drugs (Control group, CT group) (Fig. 1).

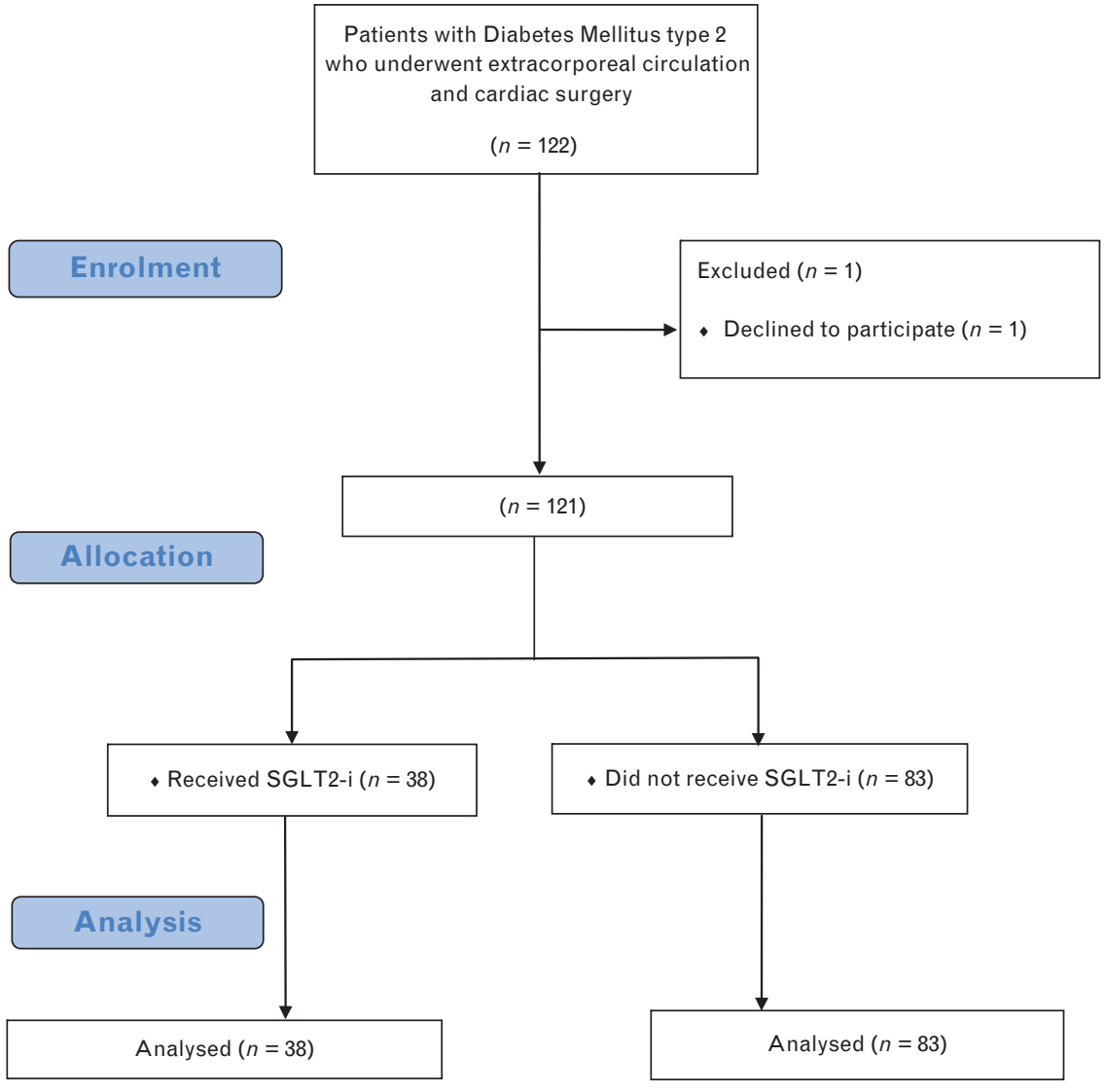
Data collection

Clinical and laboratory data were collected from the electronic patient record database. The clinical data included information on age, sex, weight, height, BMI, hypertension, New York Heart Association Functional Classification (NYHA), left ventricular ejection fraction (LVEF), Euroscore II, prescribed medications, type of surgery, time on cardiopulmonary bypass (CPB), cross-clamp time, peri-operative bleeding and diuresis, as well as the need for peri-operative transfusion, vasopressors and insulin. Net fluid balance (NFB) was calculated on the first postoperative day. The laboratory data included measurements of haemoglobin (Hb), haematocrit (Hct), serum creatinine, estimated glomerular filtration rate (eGFR) and electrolytes, as well as parameters related to acid/base balance.

Arterial blood gas analysis was conducted at several time points, including before anaesthesia induction, 10 min after the initiation of cardiopulmonary bypass (CPB),

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Fig. 1 Flow chart of diabetic patients who underwent cardiac surgery on extracorporeal circulation in Haukeland University Hospital from 1 January 2019 to 31 December 2020. All patients diagnosed with diabetes mellitus type 2.



One patient declined to participate. SGLT2-i, SGLT2-inhibitor.

and at 12-h and 24-h postsurgery. In instances of multiple measurements, the blood gas sample with the lowest base excess during the initial 12 h after surgery was recorded, with the same procedure followed for the subsequent 12 to 24 h postsurgery period.

The albumin-corrected anion gap is derived by adding 0.25 times the difference between the normal serum albumin level and the observed serum albumin level to the standard anion gap (Equation of Figge). Measurements of serum albumin were taken upon admission and on the first postoperative day. The serum albumin value at 07 : 00 on the day after surgery was used when calculating the albumin-corrected anion gap at 12 and 24 h postoperatively. This choice of time was due to the fact that substantial haemodilution occurs during the initiation of CPB, and there are no further accurate measurements of serum albumin possible.

Main outcomes

The primary aim of this study was to identify metabolic acidosis with an elevated anion gap throughout the course of cardiac surgery and the postoperative period. To exclude lactic acidosis as the underlying cause of metabolic acidosis, a correction for lactate levels was made. The corrected base excess was calculated by adding together the measured base excess (mmol l^{-1}) and the lactate level (mmol l^{-1}). By presenting both the

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uncorrected and corrected base excess values, we aimed to provide a comprehensive view of the metabolic derangements. Throughout the study period, ketone measurements in blood or urine were not obtained.

Given that patients with chronic kidney disease routinely experience chronic metabolic acidosis, we have also presented acid/base data in groups wherein exclusively patients with eGFR at least 60 ml min^{-1} m^{-2} have been included. This definition excludes patients with moderate to severe renal impairment (eGFR $<$ 60 ml min^{-1} m^{-2}), but includes those with mild kidney dysfunction (eGFR 60 to 89 ml min^{-1} m^{-2}).

Statistical analysis

Continuous variables are reported as mean with standard deviation (mean ± SD).

Categorical variables are expressed as numbers (percentage). The Shapiro–Wilk test of normality was performed for continuous variables. Two-group comparisons for continuous data were conducted using either an independent *t*-test or a Mann–Whitney Test. Categorical variables were compared using Pearson's χ^2 test or Fisher's exact test as appropriate. For multiple comparison procedure, the Bonferronis method was undertaken: statistical significance for four comparisons was indicated by *P* value of 0.0125 or less, and for two comparisons *P* value of 0.025 or less. Statistical analysis was carried out using SPSS statistics for windows, version 29.0, IBM Corp 2022.

Multivariable linear regression analysis was performed to evaluate the association between BE plus lactate and several factors as independent variables. The following candidate factors were included in the model: SGLT2-i, age, sex, prescribed metformin, peroperative insulin-infusion, eGFR, diuresis, time on CPB, cross-clamp time. The selection of these variables was based on clinical interests and impact on base excess plus lactate as an outcome. Results are provided as coefficient estimates and their corresponding confidence intervals. Multicollinearity in multivariable linear regression model was checked using multicollinearity diagnostics by variance inflation factor (VIF). Multicollinearity was assumed if the VIF score was more than 5.[15] Diagnostic plots to assess the assumptions and performance of the regression model were evaluated. Significance level considered *P* value of 0.05 or less. Regression analyses were performed in R version 4.1.1.

Results

Table 1 presents the patients' baseline characteristics. There were no significant differences observed between the groups except for values related to Hb, Hct and serum sodium. Table 2 summarises the prescribed medications: again, there were no significant differences between the two groups. In the SGLT2-i group, 22 out of 38 (58%) of the patients used empagliflozin, 14 (37%) used dapagliflozin, one used canagliflozin and one used ertugliflozin. Nine patients continued taking SGLT2-i within 24 h of surgery, 19 stopped 48 h before surgery, five patients stopped 72 h before surgery and one patient had stopped 96 h before surgery. Four data points were missing.

Table 1 Patient characteristics

Parameter	SGLT2-i *n* = 38	CT *n* = 83	*P*
Age (years)	66.5 ± 8.9	67.1 ± 8.2	0.726
Sex			
Male	32 (84)	66 (80)	
Female	6 (16)	17 (20)	0.624[a]
Weight (kg)	87.5 ± 20.6	87.2 ± 15.7	0.924
Height (cm)	172.9 ± 8.7	175.1 ± 8.4	0.179
BMI (kg m^{-2})	29.4 ± 6.5	28.3 ± 4.3	0.298
Haemoglobin (g dl^{-1})	14.4 ± 1.79	13.1 ± 1.78	< 0.001
Haematocrit	0.44 ± 0.05	0.40 ± 0.05	< 0.001
Pre-operative serum sodium (mmol l^{-1})	140.4 ± 1.84	139.2 ± 2.58	0.011
Pre-operative serum creatinine (μmol l^{-1})	85.8 ± 22.4	95.2 ± 63.1	0.375
Pre-operative insulin use	10 (26.3)	23 (27.7)	0.873[a]
Pre-operative eGFR (l min^{-1} m^{-2})	77.5 ± 20.0	77.3 ± 25.0	0.964
Hypertension	31 (82)	67 (81)	0.911[a]
LVEF < 50%	n = 11	n = 8	
	42.2 ± 6.7	40.6 ± 4.2	0.173
NYHA	2.3 ± 0.5	2.2 ± 0.4	0.062
Euroscore II	2.68 ± 5.89	2.58 ± 4.48	0.924

Data are mean ± SD, *n* or *n* (%). eGFR, estimated glomerular filtration rate; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association functional classification. [a]Fischer's exact test/Pearson χ^2.

Table 3 displays peri-operative data. No differences were observed between the groups in terms of the type of surgery performed, time on CPB, cross-clamp time, perioperative bleeding, diuresis or the use of intra-operative transfusions or continuous infusions of vasopressors. A combination procedure was performed in 24% of the SGLT2-i group and 17% of the CT group ($P = 0.375$), resulting in a total of 46 procedures in the SGLT2-i group and 98 procedures in the CT group. There were no off-pump procedures in either group. In the CT group, there were four re-do procedures.

A significant difference was observed between the groups in terms of the use of peroperative insulin-infusion. In the

Table 2 Use of medication

Medication	SGLT2-i *n* = 38	CT *n* = 83	*P*
Insulin	10 (26)	23 (28)	0.873
Metformin	24 (63)	60 (72)	0.312
DPP4i/GLP1/Sulfonylurea	16 (42)	27 (33)	0.327
ASA	34 (89)	72 (87)	0.673
Statins	36 (95)	74 (89)	0.322
β-blockers	27 (71)	53 (64)	0.438
ACE inhibitor/ARB	18 (47)	41 (49)	0.836
Calcium blockers	19 (50)	41 (49)	0.951
Steroids	1 (2)	1 (1)	0.568

Data are *n* (%). ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ASA, acetylsalicylic acid; DPP4i, dipeptidyl peptidase-4 inhibitors; GLP1, glucagonlike peptide 1 agonist.

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Table 3 Peri-operative data

Parameter	SGLT2-i ($n = 38$)	CT ($n = 83$)	*P*
CABG	35 (92)	70 (84)	0.242[a]
AVR	6 (16)	23 (28)	0.154[a]
MVR	5 (13)	1 (1)	0.005[a]
Aortic surgery	0	4 (5)	0.169[a]
Combination procedures	9 (24)	14 (17)	0.375[a]
CPB time (min)	96.7 ± 47.7	99.6 ± 55.0	0.784
Cross-clamp time (min)	68.9 ± 42.7	68.9 ± 45.7	0.998
Peroperative bleeding (ml)	503.7 ± 234.2	611.5 ± 519.4	0.224
Peroperative diuresis (ml)	466.7 ± 225.8	482.1 ± 269.9	0.755
Peroperative transfusions (units)	1.3 ± 1.7	1.5 ± 3.2	0.648
Peroperative continous vasopressors (yes)	15 (39)	27 (33)	0.456[a]
Peroperative insulin (yes)	16 (42)	12 (14)	<0.001[a]

Data are mean ± SD or *n* (%). AVR, aortic valve replacement; CABG, coronary artery bypass graft; CPB time, time on cardiopulmonary bypass; MVR, mitral valve repair or replacement. [a]Fischer's exact test/Pearson χ^2.

SGLT2-i group, 42% of patients received insulin compared to 14% in the CT group ($P = 0.002$).

Table 4 presents blood gas analysis in the two groups. Base excess was significantly lower in the SGLT2-i group before induction of anesthesia, during CPB, as well as at the 12-h and 24-h postoperative measurements. Similar findings were observed when base excess was corrected for serum lactate. No differences were found in serum lactate between the groups. Blood glucose was significantly higher in the CT group 24 h postoperatively.

When insulin-infusion was inititated postoperatively (during the first 2 h in the ICU), blood glucose was 10.03 ± 3.51 mmol l^{-1} in the SGLT2-i group and 12.66 ± 2.20 mmol l^{-1} in the CT group ($P < 0.001$).

Base excess in the SGLT2-i group was −4.25 ± 3.06 mmol l^{-1}, and in the CT group -2.61 ± 2.77 mmol l^{-1} ($P = 0.018$). No significant difference was observed in lactate values (SGLT2-i group, 1.77 ± 2.00 mmol l^{-1} versus CT group, 1.85 ± 1.60 mmol l^{-1}, $P = 0.841$).

Serum sodium and serum chloride were significantly higher in the SGLT2-i group compared with the CT group throughout the study. Anion gap was higher in the SGLT2-i group at 12 h postoperatively, but there was no significant difference after 24 h. Hb was significantly higher in the SGLT2-i group compared to the CT group, pre-operatively, but no significant differences were observed per and postoperatively.

Table 5 summarises the acid/base parameters in the 93 patients with eGFR of 60 ml min^{-1} m^{-2} and above: 29 patients in the SGLT2-i group and 63 patients in the CT group. Base excess and base excess plus lactate were significantly lower in the SGLT2-i group per and postoperatively.

Serum chloride was significantly higher in the SGLT2-i group compared with the CT group during the study.

Upon close examination of the data, we discovered that 12 h postsurgery, 20 out of 29 patients without renal failure in the SGLT2-i group exhibited BE of -3 mmol l^{-1} or less. When corrected for serum lactate, 12 out of the 29 patients developed metabolic acidosis, that is with a corrected BE of less than -3 mmol l^{-1} with corresponding anion gap of 15.5 ± 2.56 mmol l^{-1}. Among these 12 patients, seven did not receive insulin peroperatively.

In the CT group, 16 out of 63 patients had base excess of -3 mmol l^{-1} or less. When adjusted for serum lactate levels, five of these 16 patients had a base excess of -3 mmol l^{-1} or less. There was a significant difference between the two groups ($P < 0.001$).

NFB the first postoperative day was 4329 ± 1134 ml in the SGLT2-i group and 4061 ± 1363 ml in the CT group ($P = 0.262$).

Urine output on the first postoperative day was significantly different in the two groups. In the SGLT2-i group, it was 1600 ± 640.1 ml, and in the CT group, 1271 ± 401.3 ml ($P = 0.001$). In the corresponding groups without renal failure, diuresis was 1654 ± 633.7 ml in the SGLT2-i group and 1265.2 ± 397.2 ml in the CT group ($P = 0.001$).

Table 6 summarises the results of multivariable linear regression analysis of candidate factors associated with base excess plus lactate. Cross-clamp time was removed from the multivariable regression, due to multicollinearity between time on CPB and cross clamp time (VIF >10).

In the multivariable regression analysis, the use of SGLT2-i was an independent factor associated with the lowest base excess plus lactate [coefficient: -2.259 (95% confidence interval 3.301 to -1.217), $P < 0.001$]. In addition, an increase in eGFR was associated with an increase in BE plus lactate levels [coefficient: 0.023 (95% confidence interval 0.002 to 0.045), $P = 0.034$] during the first 12 h. Diagnostic plots, including normal Q-Q Plot, residuals versus fitted values plot, scale-location plot, residuals versus leverage plot and Cook's distance plot, were examined, and no potential issues were detected.

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Table 4 Blood gas analysis in the two groups at four peri-operative stages

Pre-operative	SGLT2-i $n = 38$	CT $n = 83$	*P*
pH	7.42 ± 0.46	7.43 ± 0.05	0.569
BE (mmol l^{-1})	-0.71 ± 2.19	0.68 ± 1.45	0.003
BE + lactate (mmol l^{-1})	- 0.07 ± 1.53	0.80 ± 0.79	0.011
pCO_2 (kPa)	4.72 ± 0.59	4.97 ± 0.63	0.112
Lactate (mmol l^{-1})	0.89 ± 0.42	1.07 ± 0.43	0.165
B-Glucose (mmol l^{-1})	7.35 ± 1.53	8.37 ± 2.03	0.029
S-K (mmol l^{-1})	4.17 ± 0.51	3.98 ± 0.35	0.125
S-Albumin (g l^{-1})	43.8 ± 5.01	42.7 ± 4.75	0.263[a]
AG (mmol l^{-1})			
AG corrected for S-albumin (mmol l^{-1})			
Hb (g dl^{-1})	14.4 ± 1.79	13.1 ± 1.78	< 0.001
S-Na (mmol l^{-1})	140.4 ± 1.84	139.2 ± 2.58	0.011
S-Cl (mmol l^{-1})			
CPB	**SGLT2-i**	**CT**	***P***
pH	7.37 ± 0.05	7.39 ± 0.04	0.009
BE (mmol l^{-1})	-1.69 ± 2.58	-0.10 ± 2.07	0.002
BE + lactate (mmol l^{-1})	-0.84 ± 2.15	1.07 ± 2.17	<0.001
pCO_2 (kPa)	5.49 ± 0.63	5.38 ± 0.50	0.302
Lactate (mmol l^{-1})	1.01 ± 0.52	1.06 ± 0.41	0.635
B-Glucose (mmol l^{-1})	7.99 ± 1.93	8.13 ± 2.02	0.716
S-K (mmol l^{-1})	4.74 ± 0.89	4.67 ± 0.74	0.702
S-Albumin (g l^{-1})			
AG (mmol l^{-1})			
AG corrected for S-albumin (mmol l^{-1})			
Hb (g dl^{-1})	10.07 ± 1.58	9.60 ± 1.62	0.137
S-Na (mmol l^{-1})	136.44 ± 4.29	134.93 ± 3.21	0.006
S-Cl (mmol l^{-1})			
Postoperative 12 h	**SGLT2-i**	**CT**	***P***
pH	7.35 ± 0.05	7.36 ± 0.05	0.150
BE (mmol l^{-1})	-4.06 ± 2.35	-1.91 ± 2.73	<0.001
BE + lactate (mmol l^{-1})	-2.52 ± 2.51	-0.35 ± 2.50	0.001
pCO_2 (kPa)	5.27 ± 0.85	5.57 ± 0.75	0.063
Lactate (mmol l^{-1})	1.54 ± 1.23	1.56 ± 1.41	0.951
B-Glucose (mmol l^{-1})	9.35 ± 2.01	9.99 ± 3.05	0.239
S-K (mmol l^{-1})	4.57 ± 0.53	4.57 ± 0.55	0.945
S-Albumin (g l^{-1})			
AG (mmol l^{-1})	12.18 ± 2.50	11.13 ± 2.26	0.024[a]
AG corrected for S-albumin (mmol l^{-1})	14.34 ± 2.93	13.14 ± 2.67	0.023[a]
Hb (g dl^{-1})	10.88 ± 1.37	10.39 ± 1.34	0.058
S-Na (mmol l^{-1})	138.24 ± 2.10	137.00 ± 2.10	0.003
S-Cl (mmol l^{-1})	109.76 ± 2.54	107.55 ± 2.52	<0.001[a]
Postoperative 24 h	**SGLT2-i**	**CT**	***P***
pH	7.37 ± 0.03	7.38 ± 0.42	0.034
BE (mmol^{-1} l)	-2.05 ± 2.74	0.01 ± 3.72	0.001
BE + lactate (mmol l^{-1})	-0.67 ± 2.73	1.04 ± 2.53	0.001
pCO_2 (kPa)	5.43 ± 0.59	5.46 ± 1.10	0.882
Lactate (mmol l^{-1})	1.44 ± 0.96	1.31 ± 0.74	0.435
B-Glucose (mmol l^{-1})	7.83 ± 1.54	8.57 ± 1.82	0.009
S-K (mmol l^{-1})	4.31 ± 0.39	4.33 ± 0.44	0.777
S-Albumin (g l^{-1})	34.2 ± 3.58	34.55 ± 4.49	0.355[a]
AG (mmol l^{-1})	11.89 ± 2.17	11.01 ± 2.03	0.052[a]
AG corrected for S-albumin (mmol l^{-1})	13.99 ± 2.57	13.12 ± 2.10	0.111
Hb (g dl^{-1})	10.30 ± 1.50	10.11 ± 1.23	0.480
S-Na (mmol l^{-1})	138.32 ± 2.25	136.67 ± 2.50	<0.001
S-Cl (mmol l^{-1})	108.08 ± 2.53	106.00 ± 2.86	<0.001[a]

Data are mean ± SD. AG, anion gap; B, blood; BE, base excess; BE + lactate, BE (mmol l^{-1}) + lactate (mmol l^{-1}); Hb, haemoglobin, S-Cl, serum chloride; S-K, serum-potassium; S-Na, serum sodium. [a]Bonferroni correction significance level $P < 0.0125$.

Discussion

In summary, we found a metabolic disparity between the groups. The pH values in the two groups were similar, likely due to the use of mechanical ventilation. However, the SGLT2-i group exhibited a higher positive anion gap and lower base excess compared with the CT group. Although the difference may be small in absolute terms, it is distinctly evident. This is interpreted as diabetic patients using SGLT2 inhibitors being at an increased risk of developing ketosis during heart surgery.

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Table 5 Acid/base parameters in patients with estimated glomerular filtration rate above 60 ml min^{-1} m^{-2}

Pre-operative	SGLT2-i $n = 29$	CT $n = 63$	*P*
pH	7.42 ± 0.04	7.43 ± 0.04	0.683
BE (mmol l^{-1})	-0.64 ± 1.93	0.57 ± 1.38	0.031
BE + lactate (mmol l^{-1})	0.11 ± 1.31	0.76 ± 0.77	0.106
pCO_2 (kPa)	4.71 ± 0.50	4.96 ± 0.62	0.151
Lactate (mmol l^{-1})	0.96 ± 0.46	1.04 ± 0.44	0.599
B-Glucose (mmol l^{-1})	7.61 ± 1.48	8.50 ± 1.90	0.078
S-K (mmol l^{-1})	4.10 ± 0.51	3.94 ± 0.33	0.193
AG (mmol l^{-1})			
AG corrected for S-albumin (mmol l^{-1})			
Hb (g dl^{-1})	14.74 ± 1.58	13.36 ± 1.73	<0.001
S-Na (mmol l^{-1})	140.4 ± 1.72	139.3 ± 2.58	0.039
S-Cl (mmol l^{-1})			
CPB	**SGLT2-i**	**CT**	***P***
pH	7.36 ± 0.05	7.39 ± 0.04	0.005
BE (mmol l^{-1})	-1.70 ± 2.50	0.11 ± 1.75	<0.001
BE +lactate (mmol l^{-1})	-0.74 ± 1.94	1.29 ± 1.97	<0.001
pCO_2 (kPa)	5.51 ± 0.64	5.41 ± 0.50	0.412
Lactate (mmol l^{-1})	0.94 ± 0.36	1.09 ± 0.43	0.121
B-Glucose (mmol l^{-1})	7.99 ± 1.82	8.23 ± 2.08	0.603
S-K (mmol l^{-1})	4.63 ± 0.83	4.61 ± 0.75	0.910
AG (mmol l^{-1})			
AG corrected for S-albumin (mmol l^{-1})			
Hb (g dl^{-1})	10.51 ± 1.50	9.84 ± 1.59	0.024
S-Na (mmol l^{-1})	136.8 ± 4.6	134.8 ± 3.1	0.004
S-Cl (mmol l^{-1})			
Postoperative 12 h	**SGLT2-i**	**CT**	***P***
pH	7.36 ± 0.06	7.37 ± 0.05	0.340
BE (mmol l^{-1})	-3.93 ± 2.47	-1.52 ± 2.38	<0.001
BE + lactate (mmol l^{-1})	-2.32 ± 2.63	-0.17 ± 2.37	0.001
pCO_2 (kPa)	5.16 ± 0.81	5.55 ± 0.67	0.025
Lactate (mmol l^{-1})	1.61 ± 1.34	1.36 ± 0.76	0.249
B-Glucose (mmol l^{-1})	9.30 ± 2.27	9.72 ± 2.87	0.487
S-K (mmol l^{-1})	4.46 ± 0.46	4.47 ± 0.48	0.901
AG (mmol l^{-1})	12.52 ± 2.47	10.88 ± 1.96	0.003[a]
AG corrected for S-albumin (mmol l^{-1})	14.75 ± 2.88	12.97 ± 2.56	0.018[a]
Hb (g dl^{-1})	10.89 ± 1.50	10.48 ± 1.42	0.218
S-Na (mmol l^{-1})	138.07 ± 2.28	136.90 ± 2.14	0.043
S-Cl (mmol l^{-1})	109.24 ± 2.40	107.34 ± 2.40	0.002[a]
Postoperative 24 h	**SGLT2-i**	**CT**	***P***
pH	7.37 ± 0.03	7.38 ± 0.04	0.087
BE (mmol l^{-1})	-1.79 ± 2.88	0.12 ± 2.28	0.002
BE + lactate (mmol l^{-1})	-0.36 ± 2.80	1.39 ± 2.33	<0.001
pCO_2 (kPa)	5.47 ± 0.55	5.51 ± 2.25	0.882
Lactate (mmol l^{-1})	1.50 ± 1.03	1.24 ± 0.58	0.130
B-Glucose (mmol l^{-1})	7.82 ± 1.63	8.72 ± 1.59	0.007
S-K (mmol l^{-1})	4.24 ± 0.34	4.25 ± 0.36	0.906
AG (mmol l^{-1})	11.70 ± 2.37	11.81 ± 2.06	0.093[a]
AG corrected for S-albumin (mmol l^{-1})	13.87 ± 2.75	12.96 ± 1.97	0.185[a]
Hb (g dl^{-1})	10.54 ± 1.54	10.27 ± 1.30	0.453
S-Na (mmol l^{-1})	138.00 ± 3.03	136.56 ± 2.47	0.014
S-Cl (mmol l^{-1})	107.68 ± 2.89	105.60 ± 2.75	0.002[a]

Data are mean ± SD. AG, anion gap; BE + lactate, BE (mmol l^{-1}) + lactate (mmol l^{-1}); Hb, haemoglobin; S-Cl, serum chloride; S-K, serum-potassium; S-Na, serum sodium. . [a] Bonferroni correction significance level $P < 0.0125$.

Other factors that may contribute to decreased base excess and increased anion gap include lactic acidosis and renal failure: these have been ruled out in our study cohort. Other conceivable causes could include intoxication or liver failure, neither of which were present in the study patients. Consequently, the high anion gap must signify a state of ketosis.

Our study, using clinical data from a tertiary hospital, provides insights into managing patients on SGLT2-i during major surgery. While the cardiovascular and renal benefits of these drugs are established, their impact during surgery, especially that involving extracorporeal circulation, is less understood. By excluding patients with renal impairment, our research further clarifies the effects of SGLT2-i drugs on metabolic acidosis and ketoacidosis.

The reported incidence of eDKA varies widely across studies. For instance, in the EMPA-REG OUTCOME

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Table 6 Multivariable linear regression analysis to identify factors associated with the base excess plus lactate

Variable	Coefficient	*P*	VIF
(Intercept)	-3.741 (-9.148 to 1.665)	0.1731	Not applicable
Use of SGLT2-i	-2.259 (-3.301 to -1.217)	< 0.001	1.136
Age	0.0148 (-0.047 to 0.077)	0.634	1.136
Sex (male)	0.470 (-0.719 to 1.660)	0.434	1.057
Use of metformin	0.453 (-0.578 to 1.486)	0.385	1.097
eGFR	0.023 (0.002 to 0.045)	0.034	1.412
Peroperative insulin-infusion	-0.240 (-1.409 to 0.928)	0.684	1.179
Diuresis	-0.0007 (-0.003 to 0.001)	0.447	1.227
Time on CPB	0.0059 (-0.004 to 0.016)	0.223	1.251

Data are coefficient (95% confidence interval). The VIF provides an index that measures how much the variance of an estimated regression coefficient is increased because of collinearity. CPB, cardiopulmonary bypass; eGFR, estimated glomerular filtration rate; VIF, variance inflation factor.

trial, only 0.09% of patients taking empagliflozin developed DKA, compared to 0.04% in the placebo group, showing no significant difference.[1] In contrast, Iwasaki *et al.*[16] reported a 22.6% rate of eDKA in intensive care patients using SGLT2-i, compared to 8.1% in controls.

Murugesan *et al.*[17] reported a 71% incidence of eDKA in cardiac surgery patients on SGLT2-i, but this estimate may be inflated as they defined eDKA as any postoperative ketone level more than 0.6 mmol l^{-1}. However, Snel *et al.*[18] demonstrated that ketone levels naturally rise to 0.95 ± 0.5 mmol l^{-1} in cardiac surgery patients, with or without diabetes. This suggests that Murugesan's threshold may be too low, as transient postoperative ketonaemia is common even in nondiabetic patients.

Additionally, Auerbach *et al.*[19] reported a 15% incidence of postoperative eDKA in SGLT2-i users undergoing heart surgery, while Steinhorn *et al.*[20] observed that all patients who continued SGLT2-i developed anion gap acidosis, likely due to ketoacids. In a recent Korean study, the incidence of eDKA was found to be around 60% in cardiac surgery patients, and in our study, the incidence was approximately 40%.[21]

Our findings, supported by other studies, suggest that the risk of eDKA is significantly higher in cardiac surgery compared with other types of surgery. While Mehta *et al.*[22] reported a relatively low incidence of eDKA in nonemergency (0.17%) and emergency surgery (1.1%), our data indicate a greater risk of metabolic acidosis in cardiac surgery patients, underscoring the need for targeted peri-operative management in high-risk case.

At the time of the study, there was no standardised treatment protocol for DM2 and SGLT2-i in the department. Due to concerns about the development of DKA, the approach to treatment varied among physicians. Some used insulin-infusion as a prophylactic measure, while others relied on acid-base assessments and initiated insulin when base excess levels decreased. Some 42% of patients using SGLT2-i received insulin-infusion from the start of the procedure. Nevertheless, they exhibited lower base excess compared with the control group. There is no important difference observed between the groups regarding pre-operative medication, insulin use or risk assessment scores. Consequently, there is no reason to believe that patients in the SGLT2-i group manifest a more severe disease compared with patients in the CT group. The higher proportion of patients receiving insulin infusion in the SGLT2-i cohort could suggest a perceived need for more aggressive glycaemic control or a greater concern for DKA development. The specific criteria for diagnosing eDKA vary across different guidelines.[23,24] The diagnosis typically requires an elevated blood glucose level and increased blood ketones ($\geq$3 mmol l^{-1}), along with one of the following: elevated anion gap (>10 mmol l^{-1}) or metabolic acidosis (pH $\leq$ 7.3, base excess $\leq$-3 mmol l^{-1}, HCO_3^- $\leq$18 mmol l^{-1}). In Norway, European/British guidelines are followed, indicating blood glucose levels below 11 mmol l^{-1} to diagnose euglycaemia.

In the absence of ketone measurements, we conducted a subgroup analysis excluding patients with renal impairment; 12 patients out of 29 patients (41%) in this subgroup with SGLT2-i exhibited a base excess of -3 mmol l^{-1}or less. This indicates that 12 out of the 38 patients (32%) in the SGLT2-i group likely developed ketoacidosis. This is further confirmed by a higher albumin-corrected anion gap in the SGLT2-i group 12 h postoperatively. In SGLT2 i-associated eDKA, the anion gap is often only moderately elevated or even normal. This is typically due to bicarbonate loss in the kidney, likely caused by an indirect effect on the Na^+/H^+ exchange in the proximal tubule. As a result, patients may develop hyperchloraemic acidosis or mixed acidosis with a mild anion gap elevation. In our dataset, serum chloride levels were notably higher compared with the control group, further supporting this mechanism.[25]

Ketoacidosis is well known to occur as a result of various precipitating factors.[23]

Major surgery, in particular, cardiac surgery, is a potential trigger of eDKA because of pre-operative fasting, discontinuation of insulin, large fluid shifts and prolonged surgical procedures. Many case reports document severe eDKA occurring both peri-operatively and for an extended postoperative period.[14,26]

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In addition, the involvement of the CPB machine is known to provoke a complex systemic inflammatory response.[27]

The natural stress response includes the release of counter-regulatory hormones such as glucagon, growth hormone, catecholamines and corticosteroids, as well as the onset of acute insulin resistance and insulin deficiency.

These factors contribute to facilitating processes such as gluconeogenesis, glycogenolysis, ketogenesis and lipolysis, with a resulting increase in plasma ketone bodies and thereby a potential initiation of eDKA in SGLT2-i treated patients.[28,29]

Various potential mechanisms have been suggested to explain eDKA linked to SGLT2-i, such as diminished insulin secretion and increased plasma glucagon. A publication by Perry *et al.*[30] found that the development of eDKA in SGLT2-i treated rats relies on simultaneous dehydration and insulin deficiency for the process to take place, the 'two hit hypothesis'.

Our investigation revealed higher pre-operative concentrations of Hb and sodium in individuals receiving SGLT2-i. Following surgery, the SGLT2-i group had increased diuresis compared with the control group, while NFB and blood product supplementation remained similar in both cohorts. This resulted in greater volume depletion in the SGLT2-i group compared with the control group, and consequently supports the two-hit hypothesis.

In our study, 42% of the patients in the SGLT2-i group received insulin intra-operatively, and insulin-infusion was initiated postoperatively in an additional 10 patients, bringing the total to 26 out of 38 patients (68%) in the SGLT2-i group receiving insulin. When insulin infusion was initiated postoperatively, the blood glucose levels were significantly lower in the SGLT2-i group compared with the CT group ($P < 0.001$). This is interpreted as insulin treatment being started due to the worsening of acidosis, rather than in response to high blood glucose levels.

During the study period, there were no established guidelines for discontinuing SGLT2-i before surgery. Consequently, six out of the 38 patients who had ceased the medication well before the surgery were included. One could speculate whether excluding these patients would have led to an even more pronounced difference between the groups.

The discussion regarding cessation and restarting of SGLT2-i after episodes of eDKA is still a controversial topic. Most medical societies advocate for a cessation period of 3 to 4 days, resuming the medication only when the patient is stable and discharged from the hospital. However, an increasing number of studies are challenging the widespread recommendation to discontinue SGLT2-i drugs. Khunti *et al.*[29,31] showed favourable effects of SGLT2-i on critically ill intensive care patients, including patients with COVID 19. Dhatariya[32] argue that SGLT2-i should only be withheld for one day before surgery, and serum ketones should be controlled until a normal diet is resumed.

The positive effects of SGLT2-i are compelling. Reports indicate an elevation in plasma beta-hydroxybutyrate without evident ketoacidosis in 12 to 20% of individuals using SGLT2-i. It is hypothesised that this mild hyperketonaemia may be responsible for the cardiac and renal effects of SGLT2-i by influencing energy substrate utilisation at the mitochondrial level.[33] This is referred to as the 'thrifty substrate hypothesis' in the EMPA-REG OUTCOME trial.[34] The delicate balance between the benefits of SLGT2-i drugs and their risk of causing eDKA, especially during major surgery, necessitates careful consideration and monitoring.

After the introduction of SGLT2-i in 2013, there has been a shift in the management of DM2, from an emphasis on glycaemic control to a more comprehensive approach centred on cardiovascular and renal protection. The recent meta-analysis by Usman *et al.*[35] supports SGLT2-i efficacy for heart failure and cardiovascular diseases, with or without DM2.

The number of individuals using SGLT2-i is on the rise, and there is significant potential for additional growth driven by new indications and innovative therapeutic agents. This is further supported by the recent FDA approval of dual SGLT1 and 2 inhibitors earlier this year.[36]

Strength and limitations

The SGLT2-i group and the CT group exhibited comparable patient and clinical characteristics, indicating a well balanced study population. Moreover, both groups had similar pre-operative medical histories, including comorbidities, medication use and type of surgery. This ensures a fair comparison of outcomes between SGLT2-i users and nonusers in the context of cardiac surgery.

The single-centre design and small sample size may limit the study's generalisability and its ability to detect clinically meaningful differences. Additionally, the retrospective data collected are incomplete, as it relies on past medical records, which are not complete. HCO_3- and serum chloride were not documented pre and peri-operatively, resulting in a lack of anion gap values at the same time points. Moreover, the retrospective design hinders establishing causality, relying solely on associations. The absence of a standardised treatment protocol and varied approaches among physicians underscore the complexity of managing this population. However, the most significant limitation is the absence of measurements for ketone bodies in blood and urine. β-hydroxybutyrate, the primary ketone produced at the onset of DKA, can be easily measured.

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In today's context, within the operating theatre and the cardiac ICU, anaesthesiologists, intensivists and nurses undergo training to recognise the condition. However, the most crucial aspect is the establishment of effective ketone monitoring protocols and the education of staff on regular wards, given that the condition can also manifest pre and postoperatively.

In conclusion, our study highlights a significant risk of ketosis, potentially leading to metabolic acidosis, and eventually eDKA, in diabetic patients using SGLT2-i undergoing heart surgery. With the increasing use of SGLT2-i, it is crucial to strike a balance between the positive effects on cardiovascular and renal health, and the associated risk of eDKA, particularly in high-stress peri-operative settings. The findings advocate for further research, consensus on guidelines and enhanced medical staff education to ensure the well tolerated and effective use of SGLT2-i in diabetic patients undergoing heart surgery.

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